LEVELHULL · FEASIBILITY ASSESSMENT

LevelHull feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade), closed on the TRL 3 design

LevelHull feasibility assessment

LevelHull is feasible on paper: the physics holds with margin once the water inside is down to the lake level (LVH-STM-001, LVH-CAL-001), with the first minutes after swamping marginal until bailed (D-O2, D-O3 with Amish), every part can be made by a beach yard and a tailor with hand tools, and every material is sold in East Africa; the weak points are not technical but the life of the covers and straps in the sun, the lost space the owners must accept and an unknown approval route (LVH-REG-001 G1). The cost is over the value-engineering target by USD 360 for two canoes. Method: STANDARDS section 23.

> **Safety:** Feasible on paper does not mean safe to use. No fitted canoe goes fishing before the pull test and a swamp test in sheltered, shallow water with a safety boat and everyone in a life jacket (LVH-BLD-001, sections 5 and 6).

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design (flotation and stability)	Swamped at the design load with the water inside at the lake level: 92 mm lowest freeboard, 761 kg m per radian, reserve 2.45, 3.9 deg with all crew on one side, afloat with the largest module lost; 76 mm with 20 % more load and 5 % less lift; 57 mm with timber at 1,200 kg/m3	Medium	A sealed canoe filled to the brim sits about 24 mm up (3 mm at the low end) and is near neutral in roll until bailed (S32, S33); bailing from brim-full takes about 22 min at 50 L/min each (S34); fixes with Amish (D-O2, D-O3)	LVH-STM-001 S1 to S7, S18, S31 to S34; LVH-CAL-001
Materials	Closed-cell PE foam (under 2 % water uptake on a cited data sheet), PVC tarpaulin, polyester webbing, hardwood, A4 stainless	Medium	UV life of covers and webbing (S10) and decay at screw holes (S23) can only be shown in the field; foam data of the actual supplier to confirm	LVH-STM-001 S8 to S10, S22, S23; BOM
Manufacturing and assembly	Ripped and planed battens, knife-cut foam strips, sewn sleeves, screws into the frames; about 6.7 h at the canoe	High	Every hull differs: the gauge and calculator handle that, but frame spacing and depth must be measured; chock screws now counterbored (D-S1)	LVH-BLD-001; LVH-STM-001 S13 to S15, S24; LVH-DDR-002
Power and controls	None: passive kit	High	Not applicable	
Supply of parts	Foam sheet, tarpaulin, webbing, stainless hardware from marine, packaging and fishing suppliers; foam is 48 % of the kit	Medium	Foam price and grade near the fishery unknown (LVH-DEC-001, to confirm item 8); stainless D-rings may need importing	BOM; LVH-MKT-001
Regulatory	Not lifesaving equipment; national small vessel rules may count or govern added buoyancy	Medium	Approval route to assess with the Kenya Maritime Authority (G1)	LVH-REG-001
Users	Fishers gain a refuge; owners lose 8.4 % of the inside volume high on the sides	Medium	Acceptance to test with a BMU; option B (no inside space) is put to Amish beside A	LVH-OPT-001; LVH-URS-001

Ratings: High, Medium, Low.